

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed backend services, large scale data pipelines, and video infrastructure at Google, YouTube, Whatnot, and Microsoft. Deep command of **Go**, **Python**, **C++**, **Java**, **Kubernetes**, and **gRPC** across Linux production environments. Owned end to end delivery of systems serving billions of requests per day, partnering with product, SRE, and data teams. Currently Works on engineering platforms that accelerate publishing and research workflows at scale. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Go, Python, C++, Java, SQL, TypeScript, JavaScript, HTML, CSS

Backend Systems: gRPC, Protocol Buffers, Kafka, Pub/Sub, REST APIs, Microservices

Infrastructure: Kubernetes, GCP, Docker, Linux, Terraform, CI/CD

Data and Storage: Bigtable, Spanner, PostgreSQL, Dataflow, Beam

Practices: On call, Code Review, Mentoring, System Design, Performance Profiling

Work Experience

Staff Software Engineer, YouTube

Remote | 02/2025 - Present

- Designed a **Pub/Sub** backed event bus replacing direct RPC calls, cutting cross service coupling and enabling 6 new feature launches in one quarter.
- Architected ingestion services in **Go** and **gRPC** handling peak loads of 4 million requests per second with sub 50 millisecond p99 latency.
- Led on call rotation across 14 services, driving mean time to resolution down to 11 minutes through refined runbooks and paging heuristics.
- Migrated legacy **C++** video metadata pipelines to a **Kubernetes** based runtime, reducing compute spend by 22 percent across three regions.
- Partnered with SRE to define SLOs for search ranking infrastructure, raising availability from 99.92 to 99.985 percent over two quarters.
- Mentored 5 senior engineers on distributed systems design, reviewing architecture docs and conducting monthly deep dives on consistency tradeoffs.

Staff Software Engineer, Whatnot

Remote | 05/2024 - 02/2025

- Owned the live auction backend in **Go**, sustaining 180,000 concurrent bidders during peak weekend drops with zero data loss incidents.
- Replaced a polling layer with a **Kafka** event stream, dropping bid latency from 800 milliseconds to 120 milliseconds end to end.
- Carried on call duties for the checkout path, reducing false positive pages by 60 percent through structured logging refinements.
- Built an internal tooling service in **Python** that automated 40 hours of weekly moderation triage work.
- Authored the team's **gRPC** API style guide and enforced it via lint checks in **CI/CD**.

Staff Software Engineer/Manager, Google

Remote | 03/2015 - 04/2024

- Drove a multi year migration from monolithic **Java** services to **gRPC** microservices, improving deployment frequency from weekly to 9 times per day.
- Shipped a **Dataflow** pipeline consolidating 12 legacy ETL jobs into one **Beam** job, reducing infrastructure cost by \$1.4 million annually.
- Managed a team of 7 engineers delivering core **Spanner** backed services used by Ads, Cloud, and Maps products.
- Designed schema sharding strategies in **Bigtable** that sustained 2.5 billion writes per day during holiday traffic peaks.
- Coached 4 engineers to promotion, running quarterly growth plans tied to measurable system reliability goals.

Software Engineer, Google

Remote | 02/2011 - 02/2015

- Built search ranking features in **C++** and **Python**, contributing to a 6 percent lift in click through rate on mobile surfaces.
- Served as primary on call for the indexing pipeline, writing 30 plus postmortems and driving action item closure within sprint boundaries.
- Implemented caching layers in **Memcached** and **Bigtable** that cut backend QPS by 35 percent during experiments.
- Owned a canonical **Protocol Buffers** schema migration affecting 200 internal services with zero downtime rollout.

Software Design Engineer II, Microsoft

Remote | 10/2007 - 12/2011

- Contributed to internal build automation tooling in **Python**, reducing developer build times by 18 minutes per cycle.
- Shipped features for Windows Live Search in **C++** on **Windows** Server, processing 700 million queries per month.
- Optimized crawler throughput by 28 percent through **Linux** kernel tuning and **SQL** query refactors on **PostgreSQL**.

Software Design Engineer Intern, Microsoft

Remote | 06/2006 - 08/2006

- Fixed 11 reported bugs in internal test harnesses and added regression coverage in the **CI/CD** pipeline.
- Prototyped a debugging visualizer in **C++** adopted by the Visual Studio extensibility team.

Education

B.S. Honors

Virginia Tech | 2004 - 2007